

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project Report

Python (EDA) · SQL (Business Analysis) · Power BI (Dashboard)

Prepared by: Janhavi

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1. Project Overview

Business Problem

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management has observed shifting purchasing patterns across demographics, product categories, and sales channels, and wants to understand which factors — discounts, reviews, seasons, or payment preferences — drive consumer decisions and repeat purchases.

Overarching Business Question

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Project Approach

- Data Preparation & Modeling (Python): Clean and transform the raw dataset for analysis.
- Data Analysis (SQL): Structure the data, simulate business transactions, and query customer segments, loyalty drivers, and purchase behavior.
- Visualization & Insights (Power BI): Build an interactive dashboard highlighting key patterns and trends.
- Report & Presentation: Summarize findings and business recommendations for stakeholders.
- GitHub Repository: House all Python scripts, SQL queries, and dashboard files.

2. Dataset Summary

The dataset contains 3,900 customer transaction records across 18 original attributes, covering demographics, product details, purchase amounts, ratings, and shopping preferences.

Key Dataset Facts

Metric	Value
Total records	3,900
Original columns	18
Product categories	Clothing, Footwear, Outerwear, Accessories
Customer locations (US states)	50
Gender split	Male: 2,652 (68%) Female: 1,248 (32%)
Average purchase amount	USD 59.76
Average review rating	3.75 / 5
Customers with discount applied	43%
Subscribed customers	27%
Missing values (before cleaning)	37 (in Review Rating only)

Column Reference

Column	Description
customer_id	Unique customer identifier
age	Customer age
gender	Male / Female
item_purchased	Specific product bought
category	Product category (Clothing, Footwear, Outerwear, Accessories)
purchase_amount	Transaction value in USD
location	US state
size / color / season	Product attributes
review_rating	Customer rating (1-5)
subscription_status	Whether customer is subscribed
shipping_type	Shipping method chosen
discount_applied	Whether a discount was used
previous_purchases	Count of prior purchases by the customer
payment_method	Payment channel used
frequency_of_purchases	How often the customer typically shops

3. Exploratory Data Analysis Using Python

All data preparation was performed in Python (pandas), following the workflow below. Code and outputs are reproduced from the project notebook.

3.1 Data Loading

```
import pandas as pd

df = pd.read_csv("customer_shopping_behavior.csv")
```

The raw CSV, containing 3,900 rows and 18 columns, was loaded into a pandas DataFrame for analysis.

3.2 Initial Exploration

`df.head()` — first look at the structure of the data:

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)
1	55	Male	Blouse	Clothing	53
2	19	Male	Sweater	Clothing	64
3	50	Male	Jeans	Clothing	73
4	21	Male	Sandals	Footwear	90
5	45	Male	Blouse	Clothing	49

`df.info()` confirmed 3,900 entries across 18 columns, with only Review Rating carrying nulls (3,863 non-null out of 3,900). `df.describe(include='all')` showed an average customer age of 44, an average purchase amount of \$59.76, and an average review rating of 3.75.

3.3 Missing Data Handling

`df.isnull().sum()` identified 37 missing values, all in `review_rating`. These were imputed using the median rating within each product category, preserving category-level rating tendencies rather than applying a single global median:

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

A follow-up `df.isnull().sum()` confirmed zero missing values across all columns after imputation.

3.4 Column Standardization

Column names were lowercased and spaces replaced with underscores for SQL and BI compatibility, and the purchase amount column was renamed for clarity:

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

3.5 Feature Engineering

Two new features were engineered to support segmentation analysis:

age_group — quartile-based age segmentation

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

purchase_frequency_days — numeric equivalent of purchase frequency

```
frequency_mapping = {
    'Fortnightly': 14, 'Weekly': 7, 'Monthly': 30, 'Quarterly': 90,
    'Bi-Weekly': 14, 'Annually': 365, 'Every 3 Months': 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

3.6 Data Consistency Check

discount_applied and promo_code_used were compared to check whether they carried duplicate information:

```
(df['discount_applied'] == df['promo_code_used']).all()
# Output: True
```

Since the two columns were identical for every record, promo_code_used was dropped as redundant:

```
df = df.drop('promo_code_used', axis=1)
```

3.7 Database Integration

The cleaned DataFrame was loaded into a MySQL database (customer_behavior_db) using SQLAlchemy, to simulate a real business data warehouse for the SQL analysis phase:

```
from sqlalchemy import create_engine
from urllib.parse import quote_plus

username = "root"
password = quote_plus("*****") # credential masked for security
host = "localhost"
port = "3306"
database = "customer_behavior_db"

engine = create_engine(f"mysql+mysqlconnector://{username}:{password}@{host}:{port}/{database}")

df.to_sql("customer", engine, if_exists="replace", index=False)
```

Security note

The original notebook has the real MySQL password hardcoded in plain text. It has been masked here. Before sharing this notebook publicly (e.g. on GitHub), replace the hardcoded password with an environment variable (e.g. os.environ['DB_PASSWORD']) and rotate the exposed password.

4. Data Analysis Using SQL (Business Transactions)

The cleaned dataset produced in the Python phase was loaded into a relational database (table: customer, database: customer_behavior_db) to simulate real business transactions. Ten business questions were answered using SQL, covering revenue, discounts, ratings, shipping, subscriptions, product performance, customer segmentation, and demographics. Each query below is shown with its actual output, executed against the cleaned dataset (3,900 customer records).

1. Total revenue generated by male vs. female customers

Query

```
select gender, sum(purchase_amount) as revenue
from customer
group by gender;
```

Output

gender	revenue
Female	75191
Male	157890

2. Customers who used a discount but still spent more than the average purchase amount

Query

```
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes' and purchase_amount >= (select avg(purchase_amount) from customer);
```

Output

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

Showing first 10 of 839 matching rows.

3. Top 5 products with the highest average review rating

Query

```
select item_purchased, round(avg(cast(review_rating as decimal(10,2))),2) as "average
review rating"
from customer
group by item_purchased
order by avg(cast(review_rating as decimal (10,2))) desc
limit 5;
```

Output

item_purchased	average_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. Average purchase amount: Standard vs. Express Shipping

Query

```
select shipping_type,
round(avg(purchase_amount), 2) as avg_purchase_amount
from customer
where shipping_type in ('Standard', 'Express')
group by shipping_type;
```

Output

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

5. Do subscribed customers spend more? Avg spend & total revenue: subscribers vs. non-subscribers

Query

```
select subscription_status,
count(customer_id) as total_customers,
round(avg(purchase_amount),2) as avg_spend,
```

```
round(sum(purchase_amount),2) as total_revenue
from customer
group by subscription_status
order by avg_spend, total_revenue desc;
```

Output

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. Top 5 products with the highest percentage of discounted purchases

Query

```
select item_purchased,
round(sum(case when discount_applied = 'Yes' then 1 else 0 end) * 100.0/count(*),2) as
discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

Output

item_purchased	discount_rate
Hat	50
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer segmentation: New, Returning, Loyal (by previous purchases)

Query

```
with customer_type as(
select customer_id, previous_purchases,
case
    when previous_purchases = 1 then 'New'
    when previous_purchases between 2 and 10 then 'Returning'
    else 'Loyal'
end as customer_segment
from customer
)

select customer_segment, count(*) as 'Number of Customers'
from customer_type
group by customer_segment;
```


Output

customer_segment	number_of_customers
Loyal	3116
New	83
Returning	701

8. Top 3 most purchased products within each category

Query

```
with item_counts as(
select category, item_purchased,
count(customer_id) as total_orders,
row_number() over(partition by category order by count(customer_id) desc) as item_rank
from customer
group by category, item_purchased
)

select item_rank,category,item_purchased,total_orders
from item_counts
where item_rank <= 3;
```

Output

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Pants	171
2	Clothing	Blouse	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Are repeat buyers (>5 previous purchases) more likely to subscribe?

Query

```
select subscription_status,
count(customer_id) as repeat_buyers
from customer
where previous_purchases > 5
group by subscription_status;
```

Output

subscription_status	repeat_buyers
No	2518
Yes	958

10. Revenue contribution of each age group

Query

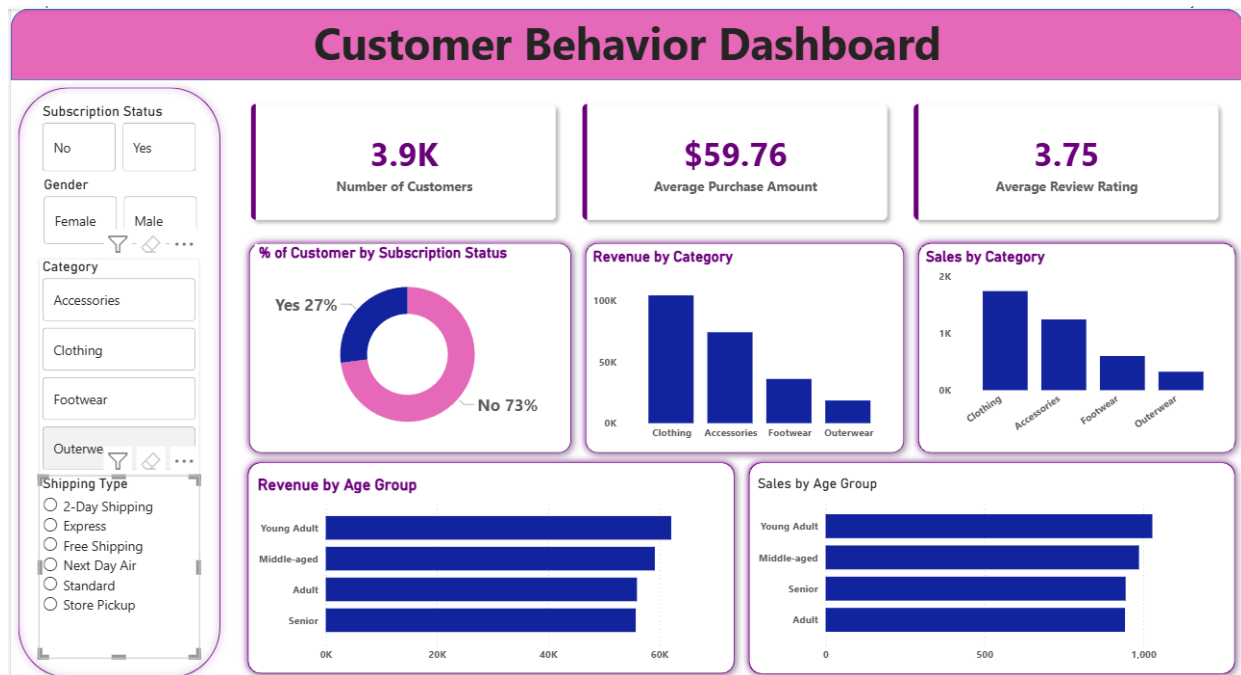
```
select age_group,  
       sum(purchase_amount) as total_revenue  
from customer  
group by age_group  
order by total_revenue desc;
```

Output

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

An interactive Power BI dashboard (Customer_Behavior_Dashboard.pbix) was built on top of the customer_behavior_db data model, giving stakeholders a single view of revenue, subscription mix, and demographic performance. The dashboard's structure, extracted from the report file, is summarized below.



Reconciliation with SQL analysis

This is the full, unfiltered dashboard view. The KPI cards (3.9K customers, \$59.76 average purchase amount, 3.75 average review rating) reconcile exactly with the SQL results in Section 4, confirming consistency between the database layer and the Power BI model. The slicers on the left (Subscription Status, Gender, Category, Shipping Type) let stakeholders drill into any of these segments interactively.

Dashboard Components (Page 1)

Visual	Title / Purpose	Fields Used
KPI Cards	Headline metrics	Number of Customers, Average Purchase Amount, Average Review Rating
Donut Chart	% of Customers by Subscription Status	subscription_status, Sum(customer_id)
Clustered Column Chart	Revenue by Category	category, Sum(purchase_amount)
Clustered Column Chart	Sales by Category	category, Sum(customer_id)
Clustered Bar Chart	Revenue by Age Group	age_group, Sum(purchase_amount)
Clustered Bar Chart	Sales by Age Group	age_group, Sum(customer_id)
Slicers	Filters for interactivity	subscription_status, gender, category, shipping_type

Dashboard Insights

- Male customers generate roughly double the revenue of female customers (\$157,890 vs. \$75,191), despite the underlying dataset skewing male (68% of records) — a demographic imbalance to account for when comparing spend by gender.
- Subscribers make up only 27% of the customer base but show comparable average spend to non-subscribers (\$59.49 vs. \$59.87), suggesting subscription is not currently driving higher basket size.
- Revenue is fairly evenly spread across the four age groups, with Young Adults contributing slightly more (\$62,143) than Seniors (\$55,763).

6. Business Recommendations

Based on the EDA, SQL analysis, and dashboard patterns, the following actions are recommended to improve engagement, retention, and revenue.

6.1 Grow the Subscription Program

- Subscribers currently spend about the same per order as non-subscribers, and make up only 27% of customers — the subscription program is not yet a strong revenue lever. Consider adding tangible perks (e.g. free shipping, early access, subscriber-only discounts) to justify subscribing and lift average order value.

6.2 Re-balance Discount Strategy

- 43% of purchases already involve a discount, and items like Hats, Sneakers, and Coats have discount rates near 50% — yet 839 customers who used a discount still spent above the average purchase amount, showing discounts aren't purely defensive. Consider tiering discounts by customer segment (e.g. lighter discounts for Loyal customers who already spend more) to protect margin.

6.3 Prioritize High-Rated, Low-Discount Products

- Gloves, Sandals, Boots, Hats, and Skirts have the highest average review ratings (3.78-3.86). Feature these prominently in marketing, since high satisfaction combined with strong demand (see Q8 top sellers per category — Jewelry, Pants, Sandals, Jackets) makes them natural upsell/cross-sell anchors.

6.4 Invest in Win-Back for the 'Returning' Segment

- The customer base splits into Loyal (3,116), Returning (701), and New (83) segments. The very small New segment suggests acquisition is slow relative to retention — marketing spend may be better allocated toward converting Returning customers to Loyal status (e.g. loyalty milestones at the 10-purchase mark) rather than broad top-of-funnel acquisition.

6.5 Shipping Type Has Limited Impact on Spend

- Average purchase amount is nearly identical between Standard (\$58.46) and Express (\$60.48) shipping. Express shipping fees are unlikely to be a meaningful upsell lever on their own — bundling Express shipping with a minimum basket threshold may be more effective than pricing it as a standalone premium.

6.6 Use Age-Group Targeting Cautiously

- Revenue by age group is close to evenly split (\$55,763-\$62,143 across four groups), meaning broad age-based segmentation alone won't meaningfully concentrate marketing spend. Combine `age_group` with category and discount-sensitivity for sharper targeting rather than relying on age alone.